

Troubleshooting Wireless Issues — Guided Notes ANSWER KEY

Unit 5.3.1 | CompTIA Network+ N10-009 Obj. 5.3 | N10-008 5.3

For instructor use / distribute after students complete the notes.

Question 1

Signal strength is measured in _____ (always negative — closer to zero is stronger). -30 dBm is excellent. -67 dBm is the minimum for reliable VoIP. Below _____ dBm, retransmissions spike and connectivity becomes unreliable.

Answer: dBm (decibels relative to one milliwatt); -80 dBm.

Question 2

Signal-to-Noise Ratio (SNR) is the difference between _____ strength and the _____ floor. A client with -40 dBm signal but a noise floor of -42 dBm has an SNR of _____ dB, which is nearly unusable despite strong signal.

Answer: Signal; noise; -2 dB (or approximately 2 dB).

Question 3

In the 2.4 GHz band, only channels _____, _____, and _____ are non-overlapping. Using channels 3 or 8 causes _____ interference that is worse than co-channel interference because the APs can't properly defer to each other.

Answer: 1; 6; 11; adjacent channel.

Question 4

A 'sticky client' stays associated with a distant AP despite a nearby AP having much stronger signal. The roaming decision is made by the _____, not the AP. Three 802.11 amendments that improve roaming are 802.11_____ (fast BSS transition), 802.11_____ (neighbor reports), and 802.11_____ (AP-directed roaming).

Answer: Client (Wi-Fi adapter/driver); 802.11r; 802.11k; 802.11v.

Question 5

An AP rated for 10 Gbps of stateful throughput might only deliver 2 Gbps with TLS inspection and IPS enabled because deeper inspection requires more _____. In a wireless context, each AP has a practical client limit — adding more clients causes _____ rather than coverage failure.

Answer: Processing power (CPU/memory); capacity degradation (airtime contention).

Question 6

A microwave oven operates at _____ GHz, directly overlapping the _____ GHz Wi-Fi band. This causes _____ interference that lasts as long as the microwave runs. The best mitigation is to steer clients to the _____ GHz band.

Answer: 2.45; 2.4; non-Wi-Fi (external RF); 5 GHz.

Question 7

On Windows, the command _____ displays the current BSSID, channel, signal strength, and radio type for the active wireless connection. This tells you which specific AP radio the client is connected to, which is useful for diagnosing _____ problems.

Answer: `netsh wlan show interfaces`; wrong AP association / sticky client / band steering.

Question 8 — Real World Application

REAL WORLD: The school auditorium has one AP on the ceiling. During regular periods it works fine. During assemblies with 300 students, the wireless network is dead — not slow, completely unresponsive. Signal strength readings from the floor are excellent at -45 dBm. Identify the problem and the correct solution.

Answer: Capacity problem, not a coverage problem. The single AP cannot serve 300 simultaneous clients — airtime contention grinds throughput to zero even though signal is strong. Solution: deploy additional APs at reduced transmit power (smaller cells, fewer clients per AP). Do not increase transmit power on the existing AP — that makes the problem worse by extending the cell without adding capacity.